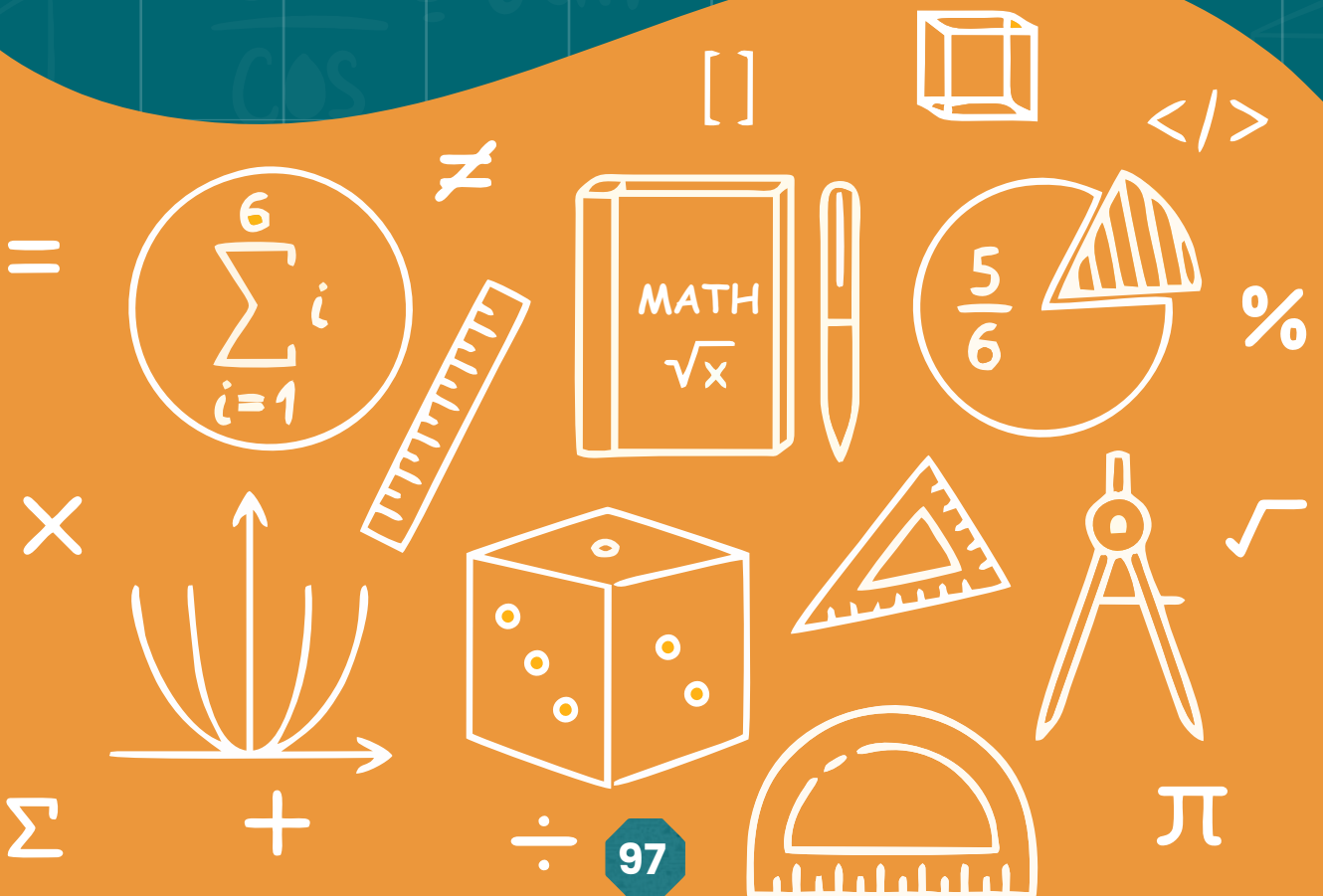


UNIT 9

AREAS AND VOLUMES



PYTHAGORAS' THEOREM

Pythagoras' Theorem is a rule in **mathematics** that applies to **right-angled triangles**. It helps us find the length of a missing side in a triangle when we know the lengths of the other two sides.

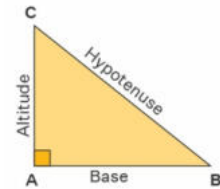
The Theorem Statement

For a **right-angled triangle**:

$$a^2 + b^2 = c^2$$

where:

- a and b are the two shorter sides (called legs) of the triangle.
- c is the **longest side**, called the **hypotenuse** (the side opposite to the right angle).



$$BC^2 = AB^2 + AC^2$$

Example Problem

Imagine we have a right-angled triangle where:

- One leg is **3 cm**
- The other leg is **4 cm**
- We need to find the **hypotenuse** (c^2).

Using Pythagoras' Theorem:

$$3^2 + 4^2 = c^2$$

$$9 + 16 = c^2$$

$$25 = c^2$$

$$c = \sqrt{25} = 5$$

So, the hypotenuse is 5 cm.

USES OF PYTHAGORAS' THEOREM

- Finding distances (like between two points on a map).
- Checking if a triangle is a right-angled triangle.
- Engineering and architecture (building structures, bridges, etc.).

Pythagorean Theorem

The sum of the squares of the sides of a right triangle equals the square of its hypotenuse.

$$a^2 + b^2 = c^2$$

The sum of the areas of the squares formed by the sides of a right triangle equals the area of the square whose side is the hypotenuse.

Activity: PQR is a right angled triangle in which $m\angle Q = 90^\circ$, $m\overline{PQ} = 6$ cm, $m\overline{QR} = 8$ cm and $m\overline{PR} = 10$ cm. Verify the Pythagoras Theorem with the above given data.

Proof: In given right angled triangle:

The measure of $\overline{QR}$, the Perpendicular is 8 cm and therefore $(\text{Perp})^2 = 8^2 = 64 \text{ cm}^2$

The measure of $\overline{PQ}$, the Base is 6 cm and therefore $(\text{Base})^2 = 6^2 = 36 \text{ cm}^2$

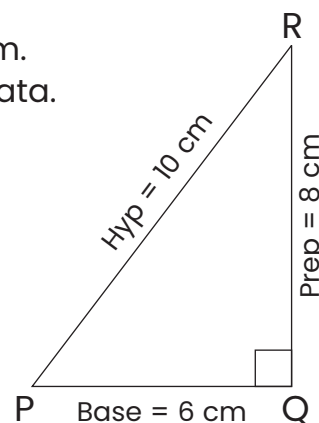
The measure of $\overline{PR}$, the Hypotenuse is 10 cm and therefore $(\text{Hyp})^2 = 10^2 = 100 \text{ cm}^2$

It is observed that: $100 = 64 + 36$

$$\text{or } (10)^2 = (8)^2 + (6)^2$$

$$\text{or } (\text{Hyp})^2 = (\text{Perp})^2 + (\text{Base})^2$$

Hence, the Pythagoras Theorem is verified.



Example 1: The sides of a triangle are 5, 12 & 13 units. Check if it has a right angle or not.

Solution: From Pythagoras Theorem, we have;

$$\text{Perpendicular}^2 + \text{Base}^2 = \text{Hypotenuse}^2$$

$$P^2 + B^2 = H^2$$

Let,

Perpendicular (P) = 12 units

Base (B) = 5 units

Hypotenuse (H) = 13 units {since it is the longest side measure}

$$\text{LHS} = P^2 + B^2$$

$$\Rightarrow 12^2 + 5^2$$

$$\Rightarrow 144 + 25$$

$$\Rightarrow 169$$

$$\text{RHS} = H^2$$

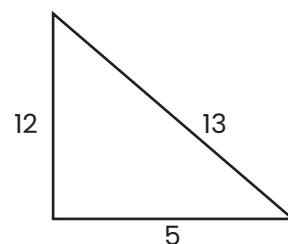
$$\Rightarrow 13^2$$

$$\Rightarrow 169$$

$$\Rightarrow 169 = 169$$

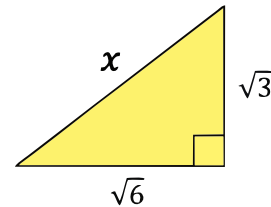
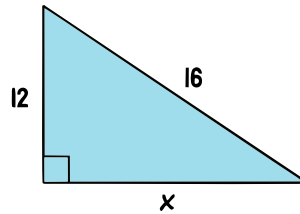
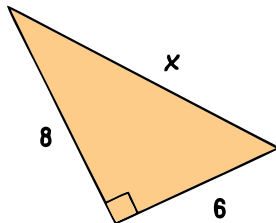
$$\text{L.H.S.} = \text{R.H.S.}$$

Therefore, the angle opposite to the 13 units side will be a right angle.



EXERCISE 9.1

Q. Find the value of x in the right triangle.



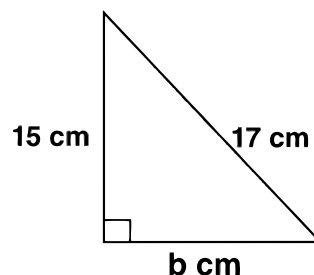
Q. The legs of a right triangle are 5 and 12 units long. What is the length of the hypotenuse?

Q. The leg of a right triangle is 8 units long and its hypotenuse is 17 units long. What is the measure of its other leg?

Q. Suppose the shorter leg of a right triangle is $2\sqrt{2}$ units. The longer leg is twice the shorter leg. Find the hypotenuse.

Q. The length of the hypotenuse of a right triangle is $4\sqrt{42}$ units. If the longest leg is half the hypotenuse, what is the length of the shortest leg?

Q. The two sides of a right-angled triangle are given as shown in the figure. Find the third side.



Example 1. Find the unknown side in right-angled triangle ABC.

Solution: Here $mAC = 5$ cm

$mBC = 5$ cm, $mAB = ?$

By Pythagoras Theorem

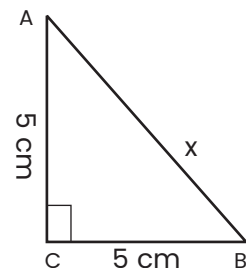
$$H^2 = P^2 + B^2$$

$$\text{or } (mAB)^2 = (mAC)^2 + (mBC)^2$$

$$\text{i.e. } x^2 = (5)^2 + (5)^2 = 25 + 25 = 50$$

$$\Rightarrow x^2 = 50 \Rightarrow x = \sqrt{50} = \sqrt{5 \times 5 \times 2}$$

$$\Rightarrow x = 5\sqrt{2} \text{ cm.}$$



Solution: Here $mAC = 1.2$ cm

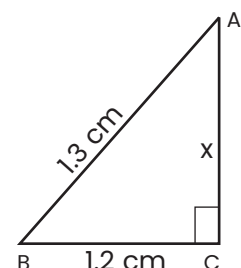
$mBC = x = ?$, $mAB = 1.3$ cm

By Pythagoras Theorem

$$H^2 = P^2 + B^2$$

$$\text{or } (mAB)^2 = (mAC)^2 + (mBC)^2$$

$$\text{i.e. } (1.3)^2 = (1.2)^2 + x^2$$



$$\begin{aligned} \Rightarrow 1.69 &= 1.44 + x^2 \\ \Rightarrow 1.69 - 1.44 &= x^2 \Rightarrow x^2 = 0.25 \\ \Rightarrow x &= \sqrt{(0.5 \times 0.5)} = 0.5 \text{ cm.} \end{aligned}$$

Example 4. An old woman takes a short cut from her home A to Tower B along a footpath AB. How much further would it be for the old woman to the Market Tower by going from A to C and then from C to B as shown in figure?

Solution: When the old woman takes a shortcut (AB) from her home at A to the Market Tower B; she completes a distance of 600 m. Again if she takes an alternative route, the distance she walk which is:

$$m_{AC} + m_{CB} = (x + 360) \text{ m}$$

Here ACB is a right angled triangle with the Hypotenuse $m_{AB} = 600 \text{ m}$

Therefore $(\text{Hyp})^2 = (\text{Perp})^2 + (\text{Base})^2$,
by Pythagoras Theorem.

$$(600)^2 = (360)^2 + (\text{Base})^2$$

$$360000 = 129600 + x^2$$

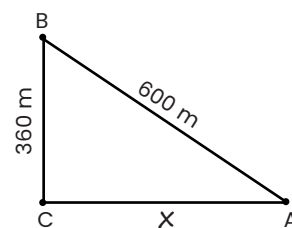
or

$$\text{Therefore } x^2 = 360000 - 129600 = 230400$$

$$\text{or } x = \sqrt{230400} = 480 \text{ m}$$

Total length of the alternative route is $360 + 480 = 840 \text{ m}$.

Thus the extra distance = $(840 - 600) \text{ m} = 240 \text{ m}$.



		480
4		230400
+4		-16
88		704
+8		-704
960		000
+0		-000
960		0

EXERCISE 9.2

- Let a , b , and c be the lengths of the sides of a triangle XYZ.
- Decide which of the following triangles are not right-angled triangles. When any of $\angle X, \angle Y$ or $\angle Z$ may be a right angle, the triangle is right-angled.

(i) $a = 5, b = 12, c = 13$

(ii) $a = 8, b = 15, c = 17$

(iii) $a = 7, b = 24, c = 25$

(iv) $a = 10, b = 24, c = 26$

(v) $a = 3, b = 4, c = 6$

(vi) $a = 4.5, b = 6, c = 7.5$

- The following are isosceles right-angled triangles. The square of the hypotenuse is given. Find the length of the congruent sides.

(i) $H^2 = 36 \text{ cm}$

(ii) $H^2 = 81 \text{ cm}$

(iii) $H^2 = 121 \text{ cm}$

(iv) $H^2 = 196 \text{ cm}$

(v) $H^2 = 3.24 \text{ cm}$

(vi) $H^2 = 8.41 \text{ cm}$

- A ladder 7.5 m long is placed against a wall. If its upper end reaches a height of 6 m along the wall, find the distance of the foot of the ladder from the wall.

- A flagpole of 15 m height is held by a supporting wire of length 25 m. Find the distance between the foot of the pole and the point where the wire meets the ground.

HERON'S FORMULA

Heron's formula is a formula to calculate the area of triangles when the three sides of the triangle. This formula is also used to find the area of the quadrilateral, by dividing the quadrilateral into two triangles, along its diagonal. If a , b and c are the three sides of a triangle, respectively, then Heron's formula is given by:

$$\text{Area of triangle using three sides} = \sqrt{s(s-a)(s-b)(s-c)}$$

Where " s " is semi-perimeter = $(a + b + c) / 2$

Example 1: Find the area of the triangle whose sides measure 10 cm, 17 cm and 21 cm. Also, determine the length of the altitude on the side which measures 17 cm.

$$s = (a+b+c) / 2 = (10+17+21) / 2 = 24$$

$$\text{Area of Triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\sqrt{24 \times 14 \times 7 \times 3}$$

$$\sqrt{7056} = 84 \text{ square cm}$$

Taking 17 cm as the base length we need to find the height

$$\text{Area, } A = 1/2 \times \text{base} \times \text{height}$$

$$1/2 \times 17 \times h = 84 \text{ or } h = 168/17 = 9.88 \text{ cm (Rounded to the nearest hundredth).}$$

Application of Heron's Formula in Finding Areas of Quadrilaterals Examples

Heron's formula is also applicable to find the area of a quadrilateral if the quadrilaterals are divided into triangular parts. Once the quadrilaterals are divided into triangular forms, apply Heron's formula to find the area of individual triangular parts. After finding the areas of each triangular part, add all the areas to find the area of quadrilateral. Now, let us understand the application of Heron's formula in finding the areas of quadrilaterals with the help of solved examples.

Example 1:

Masha has a piece of land which is shaped like a rhombus. She wants her one son and one daughter to work on the land to produce different crops. So, she has divided the land into two equal parts. Determine how much area each of them will get for their crop production if the perimeter of the land is 400 m and one of its diagonals is 160 m.

Solution:

Given: ABCD is a rhombus-shaped land.

Given that, perimeter = 400 m

Therefore, each side length of rhombus = $400/4$
= 100 cm.

Thus, we can take that AD = AB = 100 m.

Diagonal, BD = 160 m.

Now, consider the triangle ABD,

The semi perimeter of $\triangle ABD$, $s = (100 + 100 + 160)/2$

$s = 360/2$ m

$s = 180$ m.

Therefore, the semi perimeter of a triangle = 180m.

Thus, the area of triangle ABD = $\sqrt{[s(s-a)(s-b)(s-c)]}$ square units.

Now, substitute $s = 180$ m, $a = 100$, $b = 100$ and $c = 160$ in the

Heron's formula, we get

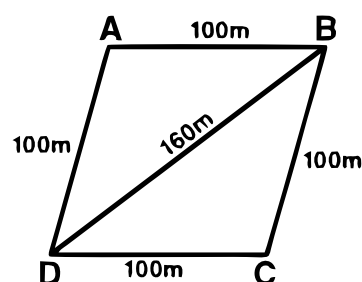
$$A = \sqrt{[180(180 - 100)(180 - 100)(180 - 160)]} \text{ m}^2$$

$$A = \sqrt{[180(80)(80)(20)]} \text{ m}^2$$

$$A = \sqrt{23040000} \text{ m}^2$$

$$A = 4800 \text{ m}^2$$

Hence, each of them will get land for their crop production of 4800 m².



EXERCISE 9.3

1. Using Heron's Formula, find the area of the following isosceles triangles, given the side lengths:

- (i) Triangle XYZ in which $x = 8$ cm, $y = 8$ cm, and $z = 10$ cm
- (ii) Triangle PQR in which $p = 14$ cm, $q = 14$ cm, and $r = 18$ cm
- (iii) Triangle ABC in which $a = 7$ cm, $b = 7$ cm, and $c = 12$ cm
- (iv) Triangle DEF in which $d = 20$ cm, $e = 20$ cm, and $f = 30$ cm

2. Find the length of the missing side of the triangle. Also, find the area of the triangle using Heron's Formula.

- (i) $a = 11$ cm, $b = 14$ cm, $S = 22$ cm, $c = \underline{\hspace{2cm}}$, and $\triangle ABC = \underline{\hspace{2cm}}$
- (ii) $d = 4.2$ cm, $e = 5.5$ cm, $S = 7.0$ cm, $f = \underline{\hspace{2cm}}$, and $\triangle DEF = \underline{\hspace{2cm}}$

Q. A triangle PQR has sides 4 cm, 13 cm and 15 cm. Find the area of the triangle.

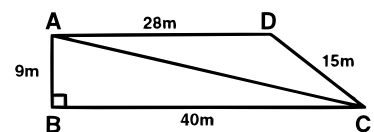
Q. Find the area of a triangle whose perimeter is 54 cm and two of its sides measure 12 cm and 25 cm.

Q. If the length of equal sides of an isosceles triangle is 5 cm and base is 6 cm, then find its area using heron's formula.

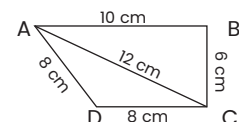
Q. The sides of a quadrilateral field, taken in order are 26 cm, 27 cm, 7 cm, 24 cm respectively. The angle contained by the last two sides is a right angle. Find its area.

Q. The school students staged a rally for a cleanliness campaign. They walked through the lane into two groups. One group walked through the lanes AB, BC and CA, and the other group walked through the lanes AC, CD and DA as shown in the figure. They cleaned the area enclosed within the lanes.

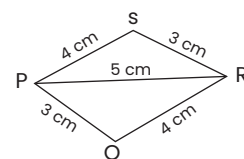
Find the total area cleaned by the students (Neglect the width of the lane), if $AB = 9$ m, $BC = 40$ m, $CD = 15$ m, $DA = 28$ m and $\angle B = 90^\circ$. Also, find which group cleaned more area and how much.



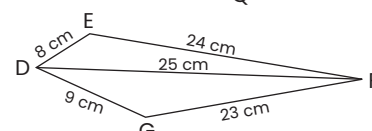
1. Find the area of quadrilateral region ABCD when $mAB = 10$ cm, $mCD = 8$ cm, $mAC = 12$ cm, $mAD = 8$ cm, and $mBC = 6$ cm



2. Find the area of quadrilateral PQRS when $mPS = mQR = 4$ cm, $mPR = 5$ cm and $mPQ = mSR = 3$ cm.

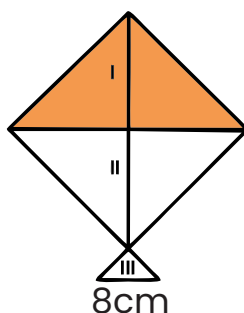


3. Find the area of quadrilateral region DEFG when $mDE = 8$ cm, $mEF = 24$ cm, $mDF = 25$ cm, $mDG = 9$ cm and $mGF = 23$ cm.



Q. A field is in the shape of a rhombus and it has green grass for 18 cows to graze. Determine how much area of grass field each cow will be getting, if each side of the rhombus is 30m and its longest diagonal is 48m.

Q. A kite is in a square shape. Its diagonal is 32 cm and its isosceles triangle of base 8 cm and side 6 cm each is made of three different shades as shown in the figure. Find how much paper of each shade has been used in it to make a kite.



SURFACE AREA AND VOLUME OF A SPHERE

Sphere

A sphere is a three-dimensional circle or we can also say that it is a circle in space. A ring and a ball have something in common.

Surface area of a sphere

The surface area of a sphere is defined as the region covered by its outer surface in three-dimensional space. A Sphere is a three-dimensional solid having a round shape, just like a circle. The formula of total surface area of a sphere in terms of pi (π) is given by:

$$\text{Surface area} = 4 \pi r^2 \text{ square units}$$

The difference between a sphere and a circle is that a circle is a two-dimensional figure or a flat shape, whereas, a sphere is a three-dimensional shape. Therefore, the area of circle is different from area of sphere.

$$\text{Area of circle} = \pi r^2$$

For any three-dimensional shapes, the area of the object can be categorized into three types.

They are:

- Curved Surface Area
- Lateral Surface Area
- Total Surface Area



Curved Surface Area: The curved surface area is the area of all the curved regions of the solid.

Lateral Surface Area: The lateral surface area is the area of all the regions except bases (i.e., top and bottom).

Total Surface Area: The total surface area is the area of all the sides, top and bottom the solid object. In case of a Sphere, it has no flat surface. Therefore, the **Total surface area of a sphere = Curved surface area of a sphere**

Example 1. Calculate the cost required to paint a football which is in the shape of a sphere having a radius of 7 cm. If the painting cost of football is Rs 2.5/square cm. (Take $\pi = 22/7$)

Solution

We know,

$$\begin{aligned}\text{The total surface area of a sphere} &= 4\pi r^2 \text{ square units} \\ &= 4 \times (22/7) \times 7 \times 7 \\ &= 616 \text{ cm}^2\end{aligned}$$

$$\text{Therefore, total cost of painting the container} = 2.5 \times 616 = \text{Rs. 1540}$$

Example 2. Calculate the curved surface area of a sphere having radius equals to 3.5 cm (Take $\pi = 22/7$)

Solution

We know,

$$\begin{aligned}\text{Curved surface area} &= \text{Total surface area} = 4\pi r^2 \\ &\text{square units} \\ &= 4 \times (22/7) \times 3.5 \times 3.5\end{aligned}$$

$$\text{Therefore, the curved surface area of a sphere} = 154 \text{ cm}^2$$

Volume of sphere

The volume of sphere is the capacity it has. It is the space occupied by the sphere. The volume of sphere is measured in cubic units, such as m^3 , cm^3 , in^3 , etc. The shape of the sphere is round and three-dimensional. It has three axes as x-axis, y-axis and z-axis which define its shape. All the things like football and basketball are examples of the sphere which have volume.

The volume here depends on the diameter or the radius of the sphere since if we take the cross-section of the sphere, it is a circle. The surface area of sphere is the area or region of its outer surface. To calculate the sphere volume, whose radius is 'r' we have the below formula:

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

Example 1: Find the volume of a sphere whose radius is 3 cm?

Solution:

Given: Radius, $r = 3 \text{ cm}$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3 \text{ cubic units}$$

$$V = \frac{4}{3} \times 3.14 \times 3^3$$

$$V = \frac{4}{3} \times 3.14 \times 3 \times 3 \times 3$$

$$V = 113.04 \text{ cm}^3$$

Example. Find the volume of sphere whose diameter is 10 cm.

Solution: Given, diameter = 10 cm

So, radius = diameter/2 = 10/2 = 5 cm

As per the formula of sphere volume, we know;

Volume = $\frac{4}{3} \pi r^3$ cubic units

$V = \frac{4}{3} \pi 5^3$

$V = \frac{4}{3} \times \frac{22}{7} \times 5 \times 5 \times 5$

$V = \frac{4}{3} \times \frac{22}{7} \times 125$

$V = 523.8$ cu.cm.

EXERCISE 9.4

Q. Find the surface area of the following spheres whose radii are given as under:

(i) 28 cm

(ii) 64 cm

(iii) 0.55 cm

(iv) 0.75 cm

(v) 1.10 cm

Q. Find the radius of each of the following spheres whose surface areas are given as under:

(i) 754 cm²

(ii) 3140 cm²

(iii) 985.6 m²

(iv) 1254.8 m²

(v) 200 m²

Q. Find the volume of the following spheres whose radii are:

(i) 25 cm

(ii) 58 cm

(iii) 3.2 m

(iv) 82 cm

(v) 4.1 m

Q. How many litres of water a sphere can contain whose radius is given as under?

(i) 85 cm

(ii) 120 cm

(iii) 3.5 m

(iv) 190 cm

(v) 2.75 m

Q. What is the volume of spheres, whose radii are:

1. 2.5 cm

2. 5 cm

3. 6 in

4. 11 m

Q. A sphere has a volume equal to 356.82 cu.cm. Find the radius of the sphere.

SURFACE AREA AND VOLUME OF A CONE

Surface area of a cone

Surface area of a cone is the complete area covered by its two surfaces, i.e., circular base area and lateral (curved) surface area. The circular base area can be calculated using area of circle formula. The lateral surface area is the side area of the cone. Let us see the formula to calculate the surface area of cone.

The surface area of a cone is the total area occupied by its surface in a 3D plane. The total surface area will be equal to the sum of its curved surface area and circular base area.

$$\text{Surface area of cone} = \pi r(r + \sqrt{h^2 + r^2})$$

where r is the radius of the circular base

h is the height of cone

Or

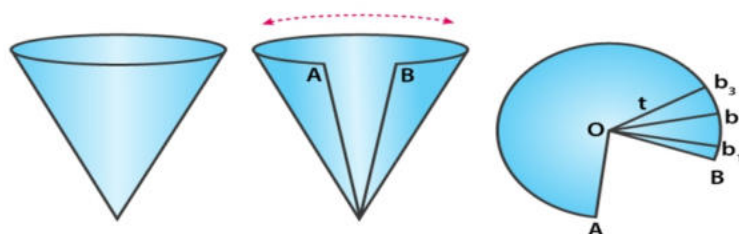
$$\text{Surface area of cone} = \pi r(r + L)$$

where L is the slant height of the cone

And

$$\text{Curved Surface area of cone} = \pi rL$$

Take a paper cone and cut it along its slant height to observe the figure being formed by the surface of the cone. Mark the two endpoints as A and B and the point of the intersection of lines as O.



If you further cut this figure into multiple pieces' viz. $Ob_1, Ob_2, Ob_3, \dots, Ob_n$ each measuring the same length as the slant height of the original cone, you will observe n triangles are formed out of it.

Now, if you try calculating the total area of this figure, you just need to add an area of these individual triangles. Hence,

$$\text{Area of figure} = \left(\frac{1}{2}\right) \times (b_1 + b_2 + b_3 + \dots + b_n) \times L$$

Since $(b_1 + b_2 + b_3 + \dots + b_n)$ is the length of the circumference $= 2\pi r$

Therefore,

$$\text{Area of figure} = \frac{1}{2} \times 2\pi \times r \times L = \pi rL$$

Hence,

The curved surface area of a cone $= \pi rL$

Total Surface Area of a Cone:

The Total surface area of a cone includes the curved surface as well as area of its base, which is given as-

$$\text{Base Area} = \pi r^2$$

$$\text{Curved Surface Area} = \pi rL$$

Thus the total surface area is given by-

$$\pi r^2 + \pi rL = \pi r(r + L)$$

Example 1. Determine the curved surface area of a cone whose base radius is 7 cm and slant height is 15 cm.

Solution: Curved surface area of a cone = πrL

$$= (22/7) \times 7 \times 15$$

$$= 330 \text{ cm}^2$$

Example 2. Calculate the Total surface area of a cone whose radius is 8 cm and height is 12 cm.

Solution. We know that the total surface area is given as:

$$\pi r(r + L)$$

Also,

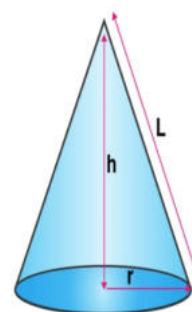
$$L = \sqrt{h^2 + r^2}$$

$$L = \sqrt{8^2 + 12^2} = \sqrt{208} = 14.42$$

$$\text{Thus, Total Surface Area} = \pi(8)(8 + 14.42)$$

$$= \pi(8)(22.42)$$

$$= 179.36\pi \text{ cm}^2$$



Volume of a Cone Formula

In general, a cone is a pyramid with a circular cross-section. A right cone is a cone with its vertex above the center of the base. It is also called right circular cone. You can easily find out the volume of a cone if you have the measurements of its height and radius and put it into a formula.

Therefore, the volume of a cone formula is given as

$$\text{The volume of a cone} = (1/3) \pi r^2 h \text{ cubic units}$$

Where,

'r' is the base radius of the cone

'h' is the height of the cone

As we can see from the above cone formula, the capacity of a cone is one-third of the capacity of the cylinder. That means if we take 1/3rd of the volume of the cylinder, we get the formula for cone volume.

Note: The formula for the volume of a regular cone or right circular cone and the oblique cone is the same.

Examples

Q.1: Calculate the volume of a cone if $r = 2$ cm and $h = 5$ cm.

Solution:

Given:

$$r = 2, h = 5$$

Using the Volume of Cone formula

The volume of a cone = $(1/3) \pi r^2 h$ cubic units

$$V = (1/3) \times 3.14 \times 2^2 \times 5$$

$$V = (1/3) \times 3.14 \times 4 \times 5$$

$$V = (1/3) \times 3.14 \times 20$$

$$V = 20.93 \text{ cm}^3$$

Therefore, the volume of a cone = 20.93 cubic units.

Q.2: If the height of a given cone is 7 cm and the diameter of the circular base is 6 cm. Then find its volume.

Solution: Diameter of the circular base = 6 cm.

$$\text{So, radius} = 6/2 = 3 \text{ cm}$$

$$\text{Height} = 7 \text{ cm}$$

By the formula of cone volume, we know;

$$V = 1/3 \pi r^2 h$$

So by putting the values of r and h , we get;

$$V = 1/3 \pi 3^2 \times 7$$

$$\text{Since } \pi = 22/7$$

Therefore,

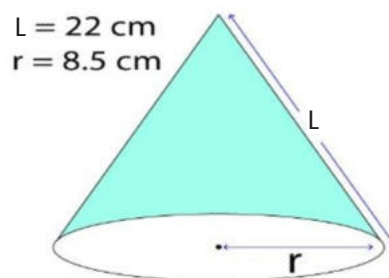
$$V = 1/3 \times 22/7 \times 3^2 \times 7$$

$$V = 66 \text{ cu.cm.}$$

EXERCISE 9.5

Q. Find the surface area of a cone with a radius of 7 cm and a slant height of 15 cm.

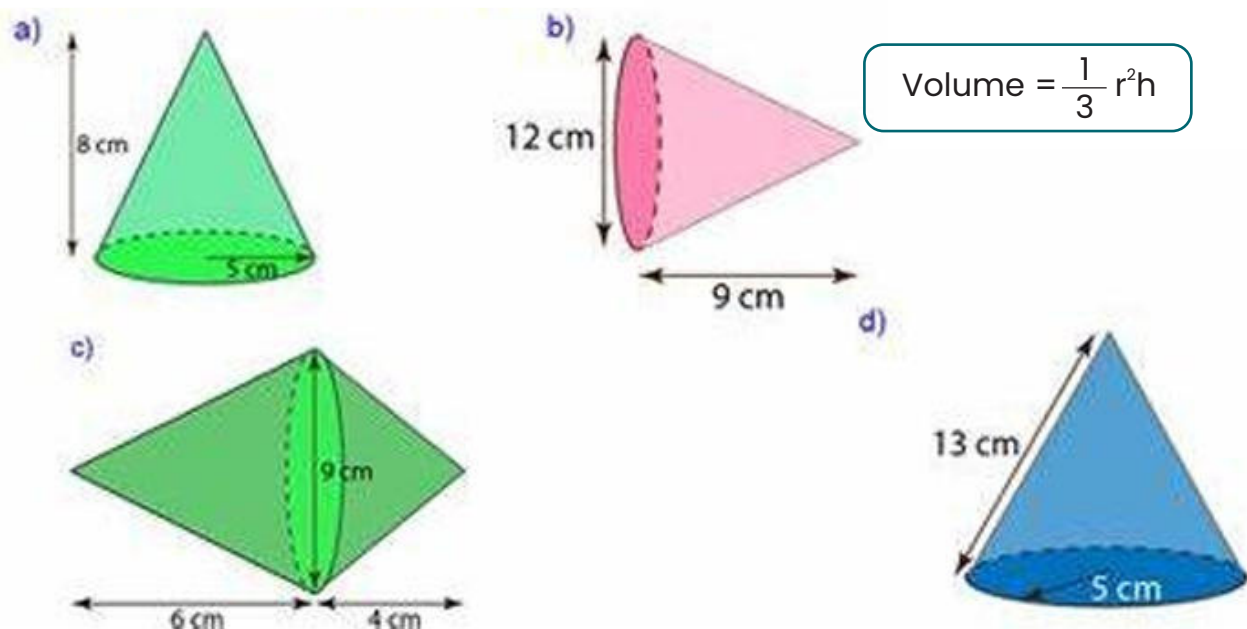
Q. Find the area of the curved surface of a cone with a radius of 8.5 cm and a slant height of 22 cm.



1. Find the unknown elements of the cone in the table given below:

S. No.	Radius r	height h	Slanting Height $s = \sqrt{h^2 + r^2}$	Base Area of Cone πr^2	Curved Surface Area of Cone $\pi r s$	Total Surface Area of Cone $\pi r(r + s)$
1.	18 cm	24 cm				
2.		48 cm	65 cm			
3.		40 cm				
4.	75 cm					28,356 cm ²
5.	90 cm	120 cm				

Calculate the volume of these cones.



1. The height of a cone is 24 cm and the diameter of its base is 14 cm. Find the volume of the cone.
2. The height and the slant height of a cone are 21 cm and 28 cm respectively. Find the volume of the cone.
3. A semicircular sheet of diameter 32 cm is folded to form a conical cup. Find the capacity of the cup.